

Daily Geometry Drill #3

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Relative Positions (4/3/2/1 tangents) · Radical Axis by subtraction · Common Chords · Orthogonal Circles ($d^2 = r_1^2 + r_2^2$).

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Compare the centre distance d with $r_1 + r_2$ and $|r_1 - r_2|$, or subtract the equations to read the radical axis.

- The circle $x^2 + y^2 - 12x - 16y + 75 = 0$ is tangent to the circle $x^2 + y^2 = 25$. Write down the number of common tangents of the two circles.
- The circles with centre $(0,0)$ radius 5 and centre $(13,0)$ radius 3 are disjoint. Write down the number of common tangents of the two circles.
- Write down the radical axis of the circles $x^2 + y^2 - 6x + 2y + 1 = 0$ and $x^2 + y^2 - 2x - 4y + 13 = 0$.
- Write down whether the circles $x^2 + y^2 = 9$ and $x^2 + y^2 + 6x + 8y + 9 = 0$ are orthogonal (yes or no).
- The circles $x^2 + y^2 = 25$ and $x^2 + y^2 - 10x + 5 = 0$ intersect in two points. Find the length of their common chord.
- Two circles have radii 3 and 5 and their centres are 4 units apart. Write down the number of common tangents.
- The circle $x^2 + y^2 - 10x - 10y + k = 0$ is tangent to both coordinate axes. Find k .
- The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. Find C .
- How many common tangents do the circles $x^2 + y^2 - 4x - 6y + 12 = 0$ and $x^2 + y^2 + 2x - 10y + 25 = 0$ have?
- Two circles have radii 7 and 3 and exactly 3 common tangents. Find the distance between their centres.

Section B Manipulation Drills

(≤50 s each)

Complete the square, locate the centres, then classify the two-circle interaction or compute the radical axis.

1. How many common tangents do the circles $x^2 + y^2 = 9$ and $x^2 + y^2 - 4x - 6y + 9 = 0$ have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4
2. The radical axis of the circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + 6x - 4y + 10 = 0$ is $2x - y + 3 = 0$. What is $a + b$?
 - A) 3
 - B) -3
 - C) 13
 - D) -13
3. The circle $x^2 + y^2 - 4x + 2y - 5 = 0$ is orthogonal to $x^2 + y^2 + 6x - 2y + k = 0$. Find k .
 - A) -9
 - B) 9
 - C) -5
 - D) 14
4. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(1, 8)$. Find the smaller possible radius.
5. Two circles of equal radius 5 have centres $(0, 0)$ and $(6, 0)$. Their radical axis has equation?
 - A) $x = 3$
 - B) $x = 6$
 - C) $y = 3$
 - D) $y = 0$
6. Two circles of radii 4 and 6 have centres 9 units apart. How many common tangents do they have?
 - A) 0
 - B) 1
 - C) 2
 - D) 3
 - E) 4

7. The circle $x^2 + y^2 = 4$ is orthogonal to $x^2 + y^2 + 8x - 6y + C = 0$. Find C .
- A) 4
B) -4
C) 25
D) 2
8. The circles $x^2 + y^2 = 36$ and $(x - 6)^2 + y^2 = 36$ intersect. Find the length of their common chord.
9. A circle is tangent to both coordinate axes in the first quadrant and passes through the point $(4, 8)$. Its possible radii are?
- A) 4 only
B) 20 only
C) 4 and 20
D) 16
10. The point $P = (1, t)$ lies on the radical axis $3x - 4y + 5 = 0$ of two circles. Find t .

Section C Substitution & Structure(≤ 75 s each)

Hard, non-routine: two or three circle tools at once — subtract for the radical axis, then turn that line into a chord or a tangency count. No brute intersection.

1. Two circles have the same radius. Their centres are $C_1(-2, 1)$ and $C_2(3, -2)$. The circles intersect in two distinct points P and Q . Which is the equation of the line PQ ? (after TMUA 2021 Paper 1 Q1)
- A) $5x + 3y = 4$
B) $3x - 5y = 4$
C) $5x - 3y = 4$
D) $5x - 3y = 1$
2. The circles $C_1 : (x - 1)^2 + y^2 = 4$ and $C_2 : (x - 5)^2 + (y - 3)^2 = 9$ have radii 2 and 3. How many common tangents do C_1 and C_2 have? (after TMUA 2019 Paper 1 Q7)
- A) 0
B) 1
C) 2
D) 3
E) 4
3. The circles $x^2 + y^2 = 169$ and $x^2 + y^2 - 6x + 8y - 119 = 0$ meet at P and Q . What is the length PQ ? (after TMUA 2016 Paper 2 Q9)
- A) 12

- B) 16
C) 24
D) 26
4. A circle in the first quadrant is tangent to both coordinate axes and passes through the point $(9, 2)$. Its two possible radii are $r_1 < r_2$. What is $r_1 + r_2$? (*adapted from JZMaths Mock 1 Q5*)
A) 14
B) 17
C) 22
D) 34
5. The circle $x^2 + y^2 + 4x - 2y + C = 0$ is orthogonal to the circle $x^2 + y^2 - 4x + 4y - 1 = 0$. What is C ? (*after Oxbridge Paper 1 Q8*)
A) -11
B) -8
C) 11
D) 5
6. A circle has equation $x^2 + ax + y^2 + by + c = 0$ where $a, b, c \neq 0$. Which condition is necessary and sufficient for the circle to be tangent to the y -axis? (*after TMUA 2021 Paper 2 Q15*)
A) $a^2 = 4c$
B) $b^2 = 4c$
C) $a^2 + b^2 = 4c$
D) $c = 0$
7. The equal circles $C_1 : (x + 2)^2 + (y - 1)^2 = 3$ and $C_2 : (x - 4)^2 + (y - 1)^2 = 3$ have centres 6 apart. A transverse common tangent with positive gradient makes an acute angle θ with the x -axis. What is $\tan \theta$? (*adapted from TMUA 2020 Paper 1 Q16*)
A) $\frac{1}{2}$
B) $\frac{\sqrt{2}}{2}$
C) $\sqrt{2}$
D) $\frac{\sqrt{3}}{3}$
8. The three circles $C_1 : x^2 + y^2 + 4x - 1 = 0$, $C_2 : x^2 + y^2 - 8y + 11 = 0$, $C_3 : x^2 + y^2 + 2x - 12y + 13 = 0$ have pairwise radical axes concurrent at R . What are the coordinates of R ? (*adapted from Vantage Mock Paper 2 Q10 / BMO1 2017 Q4*)
A) $(1, 1)$
B) $(1, -1)$

C) $(3, 1)$ D) $(1, 2)$ **Section D Challenge Ramp** $(\leq 90$ s each)

Hardest 5: combine radical axes, orthogonality and chord loops end-to-end. The numbers still cancel, but the route is hidden.

1. The circles $C_1 : (x + 2)^2 + (y - 3)^2 = 18$ and $C_2 : (x - 7)^2 + (y + 6)^2 = 2$ have centres O_1 and O_2 . What is the shortest distance between a point on C_1 and a point on C_2 ? (*adapted from TMUA 2018 Paper 1 Q3*)

A) $5\sqrt{2} - 4$ B) $5\sqrt{2} - 5$ C) $5\sqrt{2}$ D) $9\sqrt{2}$

2. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ ($r > 0$) have exactly one point in common. What is the difference between the two possible values of r ? (*after TMUA 2019 Paper 1 Q6*)

A) 8

B) 16

C) 26

D) 42

3. Three circles of radii 3, 4 and 5 are pairwise externally tangent. What is the area of the triangle formed by joining their centres? (*adapted from BMO1 2017 Q4*)

A) $12\sqrt{5}$

B) 12

C) 84

D) 30

4. Two circles of radii 3 and 4 are orthogonal. What is the length of their common chord? (*after TMUA 2016 Paper 2 Q9 / Oxbridge*)

A) $\frac{24}{5}$ B) $\frac{12}{5}$

C) 6

D) $\frac{48}{5}$

5. A point P has equal power with respect to the circles $x^2 + y^2 = 1$ and $(x - 6)^2 + y^2 = 4$. What does the locus of P describe? (*adapted from JZMaths / Vantage*)

- A) the line $x = \frac{11}{4}$
B) the circle $x^2 + y^2 = 1$
C) the y -axis
D) a single point $\left(\frac{11}{4}, 0\right)$

“Two circles are decoded by one subtraction: locate each centre, subtract the equations, then compare d with $r_1 + r_2$ and $|r_1 - r_2|$.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths